Universidad del Caribe

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CONOCIMIENTO Y CULTURA PARA EL DESARROLLO HUMANO

REPORTE DE WORDPRESS EN HA CON 3 NODOS Y UN LB
MARIADB EN HA CON 3 NODOS Y UN LB

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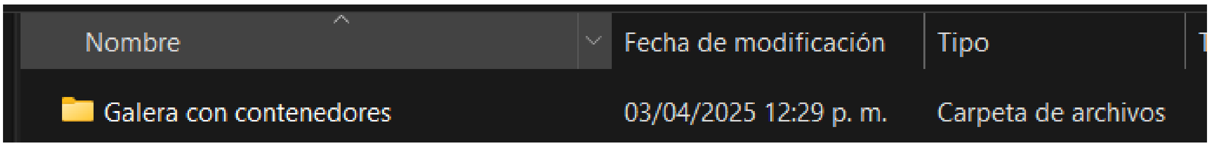
CÓMPUTO DE ALTO DESEMPEÑO

La siguiente documentación tiene como objetivo poder implementar una arquitectura HAproxy utilizando Docker para poder levantar 2 clústers, el primero es Maríadb con GaleraCluster y el otro utilizando Wordpress, balanceados por HAproxy. Se trabajó con 3 nodos para Maríadb para que este sea asíncrono por si en algún momento alguno falla, el otro nodo esté siempre disponible. Además, se levantaron 3 contenedores de wordpress para el front, que se conectan con un balanceador de carga que hace la distribución del tráfico entre ellos eficientemente.

PRIMERA PARTE (ya lo había hecho).

1.Comenzamos creando una carpeta para poder clonar ahí el repo “GALERA - DOCKER”



2. Se clona el repo de “GALERA - DOCKER”

```
C:\Documentos\Cómputo de alto desempeño\Galera con contenedores
λ git clone https://github.com/hweidner/galera-docker.git
```

3.Nos posicionamos en la carpeta que se creó al clonar el repo. “galera-docker”

```
C:\Documentos\Cómputo de alto desempeño\Galera con contenedores
λ ls
galera-docker/

C:\Documentos\Cómputo de alto desempeño\Galera con contenedores
λ cd galera-docker\

C:\Documentos\Cómputo de alto desempeño\Galera con contenedores\galera-docker (master -> origin)
λ |
```

`

4.Ahora ejecutamos el comando build para armar la imagen de galera. “docker build -t mycluster/galera .”

```
C:\Documentos\Cómputo de alto desempeño\Galera con contenedores\galera-docker (master -> origin)
λ docker build -t mycluster/galera .
[+] Building 1.8s (10/10) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 306B
 => WARN: MaintainerDeprecated: Maintainer instruction is deprecated in favor of using label (line 2)
 => [internal] load metadata for docker.io/library/mariadb:10.6
 => [auth] library/mariadb:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/4] FROM docker.io/library/mariadb:10.6@sha256:ec79aa7a81a7667885cb69b6dc0415e032f22520bd5aca77927faffca4329924
 => => resolve docker.io/library/mariadb:10.6@sha256:ec79aa7a81a7667885cb69b6dc0415e032f22520bd5aca77927faffca4329924
 => [internal] load build context
 => => transferring context: 62B
 => CACHED [2/4] RUN touch /tmp/.wsrep-new-cluster && chown -R mysql:mysql /tmp/.wsrep-new-cluster
 => CACHED [3/4] COPY galera.cnf /etc/mysql/conf.d/01-galera.cnf
 => CACHED [4/4] COPY startup.sh /startup.sh
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:657ab9c9369b1e571a88ebc95cf4158992dd60dd495dc2abe885689b30689773
 => => exporting config sha256:98eed2134b15724feabbf0ea4f1487a3b8a2b81a85981ba7239defe06cadb9e2
 => => exporting attestation manifest sha256:8b1415d89f09598556be30e7d8943bfa772ee396424dd10fb5a4cdee4980cc00
 => => exporting manifest list sha256:69b27d7472f6ae2bed976bfcdc5df4724c5bb725aec8896e13278727d0cab29c
 => => naming to docker.io/mycluster/galera:latest
 => => unpacking to docker.io/mycluster/galera:latest

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/1peyr55d6e8asvjjquitf9iyj

 2 warnings found (use docker --debug to expand):
 - JSONArgsRecommended: JSON arguments recommended for CMD to prevent unintended behavior related to OS signals (line 9)
 - MaintainerDeprecated: Maintainer instruction is deprecated in favor of using label (line 2)
```

5.Se valida que se haya creado bien la imagen galera

```
C:\Documentos\Cómputo de alto desempeño\Galera con contenedores\galera-docker (master -> origin)
λ docker images | grep galera
mycluster/galera    latest    69b27d7472f6    47 minutes ago    539MB
```

6.Se crea una nueva red para poder trabajar en ella y que no haya cruce con otro proyecto. “docker network create --subnet 172.18.0.0/16 galera”

Después se verifica si se ha creado correctamente “docker network ls”

```
C:\Documentos\Cómputo de alto desempeño\Galera con contenedores\galera-docker (master -> origin)
λ docker network create --subnet 172.18.0.0/16 galera
83fa60f91d380b661576c324b22827be53e3e19d3651ea0e2bc1baf031d69964

C:\Documentos\Cómputo de alto desempeño\Galera con contenedores\galera-docker (master -> origin)
λ docker network ls
NETWORK ID     NAME     DRIVER    SCOPE
f244dad76ec9   bridge   bridge    local
83fa60f91d38   galera   bridge    local
90c189cd74ac   host     host      local
9a804a6eb044   none     null      local
```

7.Crea una carpeta llamada “galera” y ubícate en ella.

```
C:\Documentos\Cómputo de alto desempeño\Galera con contenedores\galera-docker (master -> origin)
λ mkdir galera
```

```
C:\Documentos\Cómputo de alto desempeño\Galera con contenedores\galera-docker (master -> origin)
λ cd galera\
```

8.Se crean 3 carpetas de los 3 nodos con el comando “mkdir node{1,2,3}”.

```
C:\Documentos\Cómputo de alto desempeño\Galera con contenedores\galera-docker\galera (master -> origin)
λ mkdir node{1,2,3}
```

:

`

```
C:\Documentos\Cómputo de alto desempeño\Galera con contenedores\galera-docker\galera (master -> origin)
λ ls -la
total 4
drwxr-xr-x 1 eymar 197609 0 abr.  3 20:39 ./
drwxr-xr-x 1 eymar 197609 0 abr.  3 20:34 ../
drwxr-xr-x 1 eymar 197609 0 abr.  3 20:38 node1/
drwxr-xr-x 1 eymar 197609 0 abr.  3 20:38 node2/
drwxr-xr-x 1 eymar 197609 0 abr.  3 20:38 node3/
```

9.Nos posicionamos en cualquiera de las 3 carpetas de node (1, 2, 3).

```
C:\Documentos\Cómputo de alto desempeño\Galera con contenedores\galera-docker\galera (master -> origin)
λ cd node2\
```

10. Creamos un archivo con el nombre node1.cnf (lo mismo con 2 y 3) que contendrá lo siguiente:

"[mysqld]
wsrep_cluster_address = "gcomm://172.18.0.101,172.18.0.102,172.18.0.103"
wsrep_cluster_name = "galera-cluster"
wsrep_node_name = "node1"
wsrep_node_address = "172.18.0.101""

Solo cambiarán los campos "wsrep_node_name" y "wsrep_node_address" dependiendo en que carpeta lo estés creando, en este caso como es en la carpeta "node2" se le nombre node 2 y el ip 102 al final.

```
C:\Documentos\Cómputo de alto desempeño\Galera con contenedores\galera-docker\galera\node2 (master -> origin)
λ cat node2.cnf
[mysqld]
wsrep_cluster_address = "gcomm://172.18.0.101,172.18.0.102,172.18.0.103"
wsrep_cluster_name    = "galera-cluster"
wsrep_node_name       = "node2"
wsrep_node_address    = "172.18.0.102"
```

11.Después procedemos a darle permisos de ejecución a las carpetas

```
C:\Documentos\Cómputo de alto desempeño\Galera con contenedores\galera-docker\galera (master -> origin)
λ chown 999:999 node1

C:\Documentos\Cómputo de alto desempeño\Galera con contenedores\galera-docker\galera (master -> origin)
λ chown 999:999 node2

C:\Documentos\Cómputo de alto desempeño\Galera con contenedores\galera-docker\galera (master -> origin)
λ chown 999:999 node3
```

12.Levantamos el primer nodo con el siguiente comando "docker run -d --restart=unless-stopped --net galera --name node1 -h node1 --ip 172.18.0.101 -p 3311:3306 -v /galera/node1.cnf:/etc/mysql/conf.d/galera.cnf -v /galera/node1:/var/lib/mysql -e MYSQL_ROOT_PASSWORD=secret_galera_password -e GALERA_NEW_CLUSTER=1 mycluster/galera"

```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/galera-docker# docker run -d --restart=unless-stopped --n
et galera       --name node1 -h node1 --ip 172.18.0.101         -p 3311:3306    -v ${PWD}/galera/node1/node1.cnf:/etc/my
sql/conf.d/galera.cnf   -v ${PWD}/galera/node1:/var/lib/mysql   -e MYSQL_ROOT_PASSWORD=secret_galera_password   -e GALER
A_NEW_CLUSTER=1         mycluster/galera
e7a88708cb680e3e2aefaa1883adf401ef7d856b2cb02584e5b20b47bac03c85
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/galera-docker# docker ps
CONTAINER ID   IMAGE              COMMAND                  CREATED         STATUS         PORTS
                NAMES
e7a88708cb68   mycluster/galera   "docker-entrypoint.s…"   5 seconds ago   Up 4 seconds   0.0.0.0:3311->3306/tcp, [::]:3
311->3306/tcp   node1
```

:

13. Entramos a mysql para probar el nodo1

```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/galera-docker# mysql -h 127.0.0.1 -u root -P 3311 -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 5.5.5-10.6.21-MariaDB-ubu2004 mariadb.org binary distribution

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
4 rows in set (0,01 sec)
```

14. Levantamos el nodo2

```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/galera-docker# docker run -d --restart=unless-stopped --n
et galera \
        --name node2 -h node2 --ip 172.18.0.102 \
        -p 3312:3306 \
        -v ${PWD}/galera/node2/node2.cnf:/etc/mysql/conf.d/galera.cnf \
        -v ${PWD}/galera/node2:/var/lib/mysql \
        -e MYSQL_ALLOW_EMPTY_PASSWORD=1 \
        mycluster/galera
be2579d286fa61cd05a0f1f09536da4213be2aae0b70c18c1c5bc58f71332ad2
```

15. Verificamos si se levantó el nodo2

```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/galera-docker# docker ps
CONTAINER ID   IMAGE              COMMAND                  CREATED          STATUS          PORTS
                  NAMES
be2579d286fa   mycluster/galera   "docker-entrypoint.s…"   13 seconds ago   Up 13 seconds   0.0.0.0:3312->3306/tcp, [::]
:3312->3306/tcp   node2
e7a88708cb68   mycluster/galera   "docker-entrypoint.s…"   20 minutes ago   Up 20 minutes   0.0.0.0:3311->3306/tcp, [::]
:3311->3306/tcp   node1
```

16. Entramos de nuevo a nuestro nodo1 en mysql para crear una BD llamada “PRUEBA”

```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/galera-docker# mysql -h 127.0.0.1 -u root -P 3311 -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 5.5.5-10.6.21-MariaDB-ubu2004 mariadb.org binary distribution

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create database PRUEBA;
Query OK, 1 row affected (0,05 sec)
```

17. Comprobamos si está creada

```
mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| PRUEBA             |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0,03 sec)
```

18. Entramos a el nodo 2 y verificamos si está replicada la base de datos

```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/galera-docker# mysql -h 127.0.0.1 -u root -P 3312 -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 5.5.5-10.6.21-MariaDB-ubu2004 mariadb.org binary distribution

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| PRUEBA             |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0,02 sec)
```

19. Levantamos el nodo3

```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/galera-docker# docker run -d --restart=unless-stopped --n
et galera \
        --name node3 -h node3 --ip 172.18.0.103 \
        -p 3313:3306 \
        -v ${PWD}/galera/node3/node3.cnf:/etc/mysql/conf.d/galera.cnf \
        -v ${PWD}/galera/node3:/var/lib/mysql \
        -e MYSQL_ALLOW_EMPTY_PASSWORD=1 \
        mycluster/galera
600e58e8f1e3a572627b95d2c0491b27c3e0641be436b5749812d5d880b7a54c
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/galera-docker# docker ps
CONTAINER ID   IMAGE              COMMAND                  CREATED          STATUS          PORTS
                  NAMES
600e58e8f1e3   mycluster/galera   "docker-entrypoint.s…"   5 seconds ago    Up 4 seconds    0.0.0.0:3313->3306/tcp, [::]
:3313->3306/tcp   node3
be2579d286fa   mycluster/galera   "docker-entrypoint.s…"   10 minutes ago   Up 10 minutes   0.0.0.0:3312->3306/tcp, [::]
:3312->3306/tcp   node2
e7a88708cb68   mycluster/galera   "docker-entrypoint.s…"   31 minutes ago   Up 31 minutes   0.0.0.0:3311->3306/tcp, [::]
:3311->3306/tcp   node1
```

`

20. Verificamos si está el respaldo de la BD

```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/galera-docker# mysql -h 127.0.0.1 -u root -P 3313 -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 5.5.5-10.6.21-MariaDB-ubu2004 mariadb.org binary distribution

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| PRUEBA             |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0,00 sec)
```

:

CREACIÓN DE UN CLÚSTER CON 3 IMÁGENES EN WORDPRESS, 3 NODOS BD Y UN BALANCEADOR DE CARGA Y HAPROXY

1. Clonar el repo haproxy.

```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño# cd docker-wordpress/
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress# git clone https://github.com/tuxtter/ha
proxylb.git
Clonando en 'haproxylb'...
remote: Enumerating objects: 84, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (7/7), done.
remote: Total 84 (delta 2), reused 8 (delta 2), pack-reused 75 (from 1)
Recibiendo objetos: 100% (84/84), 22.92 KiB | 192.00 KiB/s, listo.
Resolviendo deltas: 100% (43/43), listo.
```

2.Borro la red que ya estaba ocupando ("Galera") para poder utilizarlo en este proyecto

```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño# docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
e8029650bc8b   bridge    bridge    local
29aa2aa9c2ab   galera    bridge    local
f2049b624489   host      host      local
4f838cf90464   none      null      local
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño# docker network rm galera
galera
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño# docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
e8029650bc8b   bridge    bridge    local
f2049b624489   host      host      local
4f838cf90464   none      null      local
```

3.Crear como estba anteriormente las carpetas node 1, 2 y 3

```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized/galera# mkdir -p
 node{1,2,3}
```

4. Mover los archivos node1, 2 y 3 .cnf a sus respectivas carpetas

```
total 32
drwxr-xr-x 5 root root 4096 abr  5 23:43 .
drwxr-xr-x 3 root root 4096 abr  5 23:21 ..
drwxr-xr-x 2 root root 4096 abr  5 23:43 node1
-rw-r--r-- 1 root root  193 abr  5 23:16 node1.cnf
drwxr-xr-x 2 root root 4096 abr  5 23:43 node2
-rw-r--r-- 1 root root  193 abr  5 23:16 node2.cnf
drwxr-xr-x 2 root root 4096 abr  5 23:43 node3
-rw-r--r-- 1 root root  193 abr  5 23:16 node3.cnf
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized/galera# mv node1
.cnf node1
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized/galera# ls -la
total 28
drwxr-xr-x 5 root root 4096 abr  5 23:44 .
drwxr-xr-x 3 root root 4096 abr  5 23:21 ..
drwxr-xr-x 2 root root 4096 abr  5 23:44 node1
drwxr-xr-x 2 root root 4096 abr  5 23:43 node2
-rw-r--r-- 1 root root  193 abr  5 23:16 node2.cnf
drwxr-xr-x 2 root root 4096 abr  5 23:43 node3
-rw-r--r-- 1 root root  193 abr  5 23:16 node3.cnf
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized/galera# mv node2
.cnf node2
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized/galera# mv node3
.cnf node3
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized/galera# ls -la
total 20
drwxr-xr-x 5 root root 4096 abr  5 23:44 .
drwxr-xr-x 3 root root 4096 abr  5 23:21 ..
drwxr-xr-x 2 root root 4096 abr  5 23:44 node1
drwxr-xr-x 2 root root 4096 abr  5 23:44 node2
drwxr-xr-x 2 root root 4096 abr  5 23:44 node3
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized/galera#
```

`

5. Se les da permiso

```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized/galera# chown 99
9:999 node{1,2,3}
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized/galera#
```

6. Levantamos el docker-compose

```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized# Creating dbnode
1 ...
Creating dbnode2 ...
Creating dbnode1 ... done
dbnode3      | 2025-04-07 04:09:47+00:00 [Note] [Entrypoint]: Entrypoint script for MariaDB Server 1:10.6.21+maria~ubu200
4 started.
Creating dbnode2 ... done
4 started.
webnode1 is up-to-date
webnode2 is up-to-date
webnode3 is up-to-date
master is up-to-date
Attaching to dbnode3, dbnode1, dbnode2, webnode1, webnode2, webnode3, master
dbnode2      | 2025-04-07 04:09:48+00:00 [Note] [Entrypoint]: Entrypoint script for MariaDB Server 1:10.6.21+maria~ubu200
4 started.
dbnode1      | 2025-04-07 04:09:47+00:00 [Note] [Entrypoint]: Entrypoint script for MariaDB Server 1:10.6.21+maria~ubu200
4 started.
dbnode3      | 2025-04-07 04:09:47+00:00 [Note] [Entrypoint]: Entrypoint script for MariaDB Server 1:10.6.21+maria~ubu200
4 started.
dbnode2      | 2025-04-07 04:09:48+00:00 [Note] [Entrypoint]: Entrypoint script for MariaDB Server 1:10.6.21+maria~ubu200
```

```
Papelera tualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized# docker ps -a
        R ID   IMAGE              COMMAND                  CREATED          STATUS          PORTS
           NAMES
aae3c59b2dd   mycluster/galera   "docker-entrypoint.s…"   59 seconds ago   Up 58 seconds   3306/tcp
           dbnode3
8e2d13f42f8   mycluster/galera   "docker-entrypoint.s…"   59 seconds ago   Up 58 seconds   3306/tcp
           dbnode2
d9419c31fbf   mycluster/galera   "docker-entrypoint.s…"   59 seconds ago   Up 58 seconds   3306/tcp
           dbnode1
d2f6c740b8c   haproxy            "docker-entrypoint.s…"   11 minutes ago   Up 11 minutes   0.0.0.0:80->80/tcp, [::]:80-
80/tcp     master
2a3e9154e44   wordpress          "docker-entrypoint.s…"   11 minutes ago   Up 11 minutes   80/tcp
           webnode3
d462af281c5   wordpress          "docker-entrypoint.s…"   11 minutes ago   Up 11 minutes   80/tcp
           webnode2
1c898db3229   wordpress          "docker-entrypoint.s…"   11 minutes ago   Up 11 minutes   80/tcp
           webnode1
```

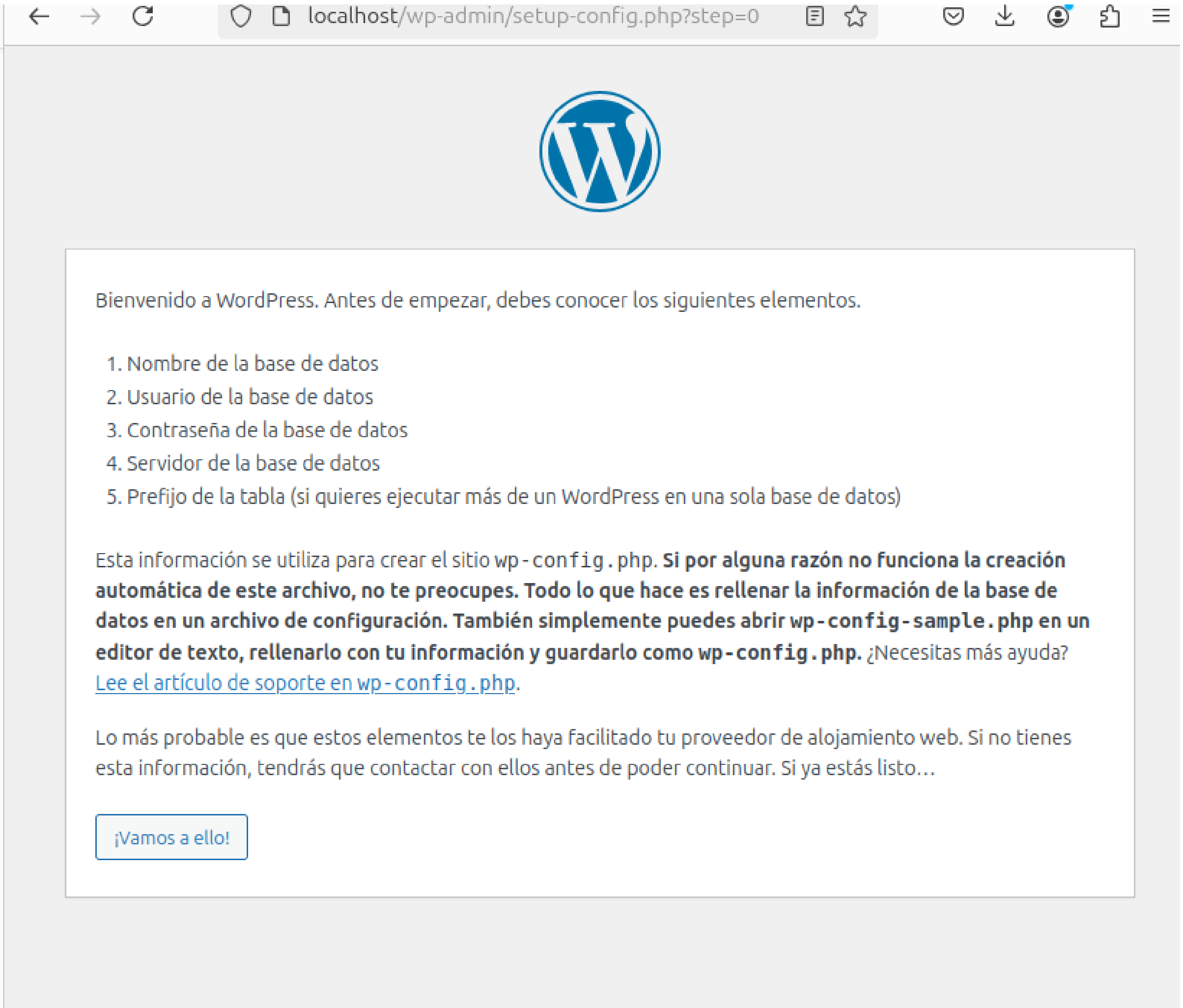
7.Se valida la creación de los proxys en los puertos 80 del Docker-compose

```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized# netstat -tlpn
Conexiones activas de Internet (solo servidores)
Proto  Recib Enviad Dirección local          Dirección remota       Estado       PID/Program name
tcp        0      0 0.0.0.0:80              0.0.0.0:*               ESCUCHAR     5583/docker-proxy
tcp        0      0 127.0.0.1:631           0.0.0.0:*               ESCUCHAR     1089/cupsd
tcp        0      0 127.0.0.53:53           0.0.0.0:*               ESCUCHAR     451/systemd-resolve
tcp        0      0 127.0.0.54:53           0.0.0.0:*               ESCUCHAR     451/systemd-resolve
tcp6       0      0 :::80                   :::*                    ESCUCHAR     5588/docker-proxy
tcp6       0      0 :::22                   :::*                    ESCUCHAR     1/init
tcp6       0      0 ::1:631                 :::*                    ESCUCHAR     1089/cupsd
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized#
```

:

`

8.Verificamos en el navegador si aparece el asistente de instalación de wordpress en local host



9. Se valida el balanceador de carga usando curl, se ve en que webnode está trabajando, esto quiere decir que nuestro loadbalancer está trabajando bien.

```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized# curl localhost
webnode1     | 172.18.0.110 - - [07/Apr/2025:04:21:09 +0000] "GET / HTTP/1.1" 302 230 "-" "curl/8.5.0"
webnode1     | 172.18.0.110 - - [07/Apr/2025:04:21:09 +0000] "GET / HTTP/1.1" 302 230 "-" "curl/8.5.0"
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized# curl localhost
webnode2     | 172.18.0.110 - - [07/Apr/2025:04:21:17 +0000] "GET / HTTP/1.1" 302 230 "-" "curl/8.5.0"
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized# webnode2     | 1
72.18.0.110 - - [07/Apr/2025:04:21:17 +0000] "GET / HTTP/1.1" 302 230 "-" "curl/8.5.0"
curl localhost
webnode3     | 172.18.0.110 - - [07/Apr/2025:04:21:18 +0000] "GET / HTTP/1.1" 302 230 "-" "curl/8.5.0"
webnode3     | 172.18.0.110 - - [07/Apr/2025:04:21:18 +0000] "GET / HTTP/1.1" 302 230 "-" "curl/8.5.0"
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized# curl localhost
webnode1     | 172.18.0.110 - - [07/Apr/2025:04:21:19 +0000] "GET / HTTP/1.1" 302 230 "-" "curl/8.5.0"
webnode1     | 172.18.0.110 - - [07/Apr/2025:04:21:19 +0000] "GET / HTTP/1.1" 302 230 "-" "curl/8.5.0"
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized# curl localhost
webnode2     | 172.18.0.110 - - [07/Apr/2025:04:21:20 +0000] "GET / HTTP/1.1" 302 230 "-" "curl/8.5.0"
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized# webnode2     | 1
72.18.0.110 - - [07/Apr/2025:04:21:20 +0000] "GET / HTTP/1.1" 302 230 "-" "curl/8.5.0"
curl localhost
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized# webnode3     | 1
72.18.0.110 - - [07/Apr/2025:04:21:24 +0000] "GET / HTTP/1.1" 302 230 "-" "curl/8.5.0"
webnode3     | 172.18.0.110 - - [07/Apr/2025:04:21:24 +0000] "GET / HTTP/1.1" 302 230 "-" "curl/8.5.0"
```

:

`

10.Entramos a un nodo de bd.

```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized# docker exec -it
 dbnode2 bash
mysql@dbnode2:/$ mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 8
Server version: 10.6.21-MariaDB-ubu2004 mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> show database;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server versi
on for the right syntax to use near 'database' at line 1
MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
```

11. Creamos la bd de wordpress en el nodo2

```
MariaDB [(none)]> create database wordpress;
Query OK, 1 row affected (0.062 sec)

MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
| wordpress          |
+--------------------+
5 rows in set (0.002 sec)

MariaDB [(none)]>
```

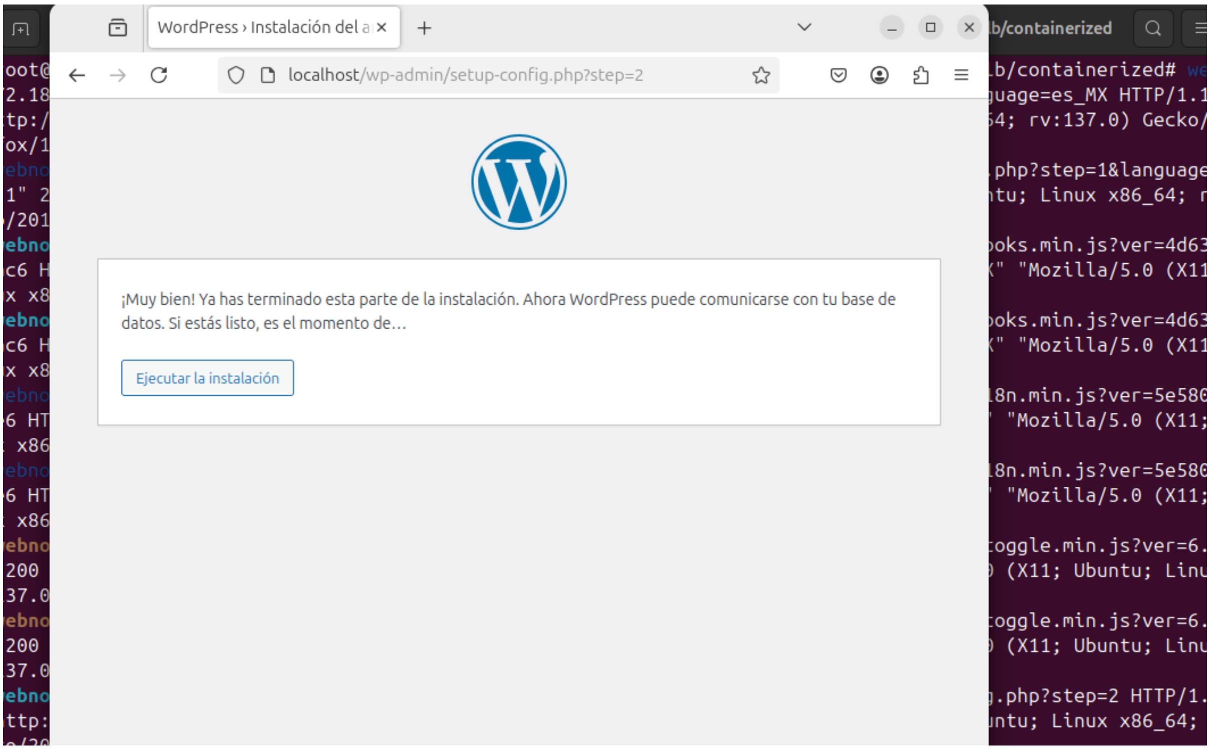
12.Entramos al nodo3 en la bd y verificamos la visualización de la bd wordpress creada en el nodo2

```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized# docker exec -it
 dbnode3 bash
mysql@dbnode3:/$ mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 8
Server version: 10.6.21-MariaDB-ubu2004 mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> show databses;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server versi
on for the right syntax to use near 'databses' at line 1
MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
| wordpress          |
+--------------------+
5 rows in set (0.016 sec)
```

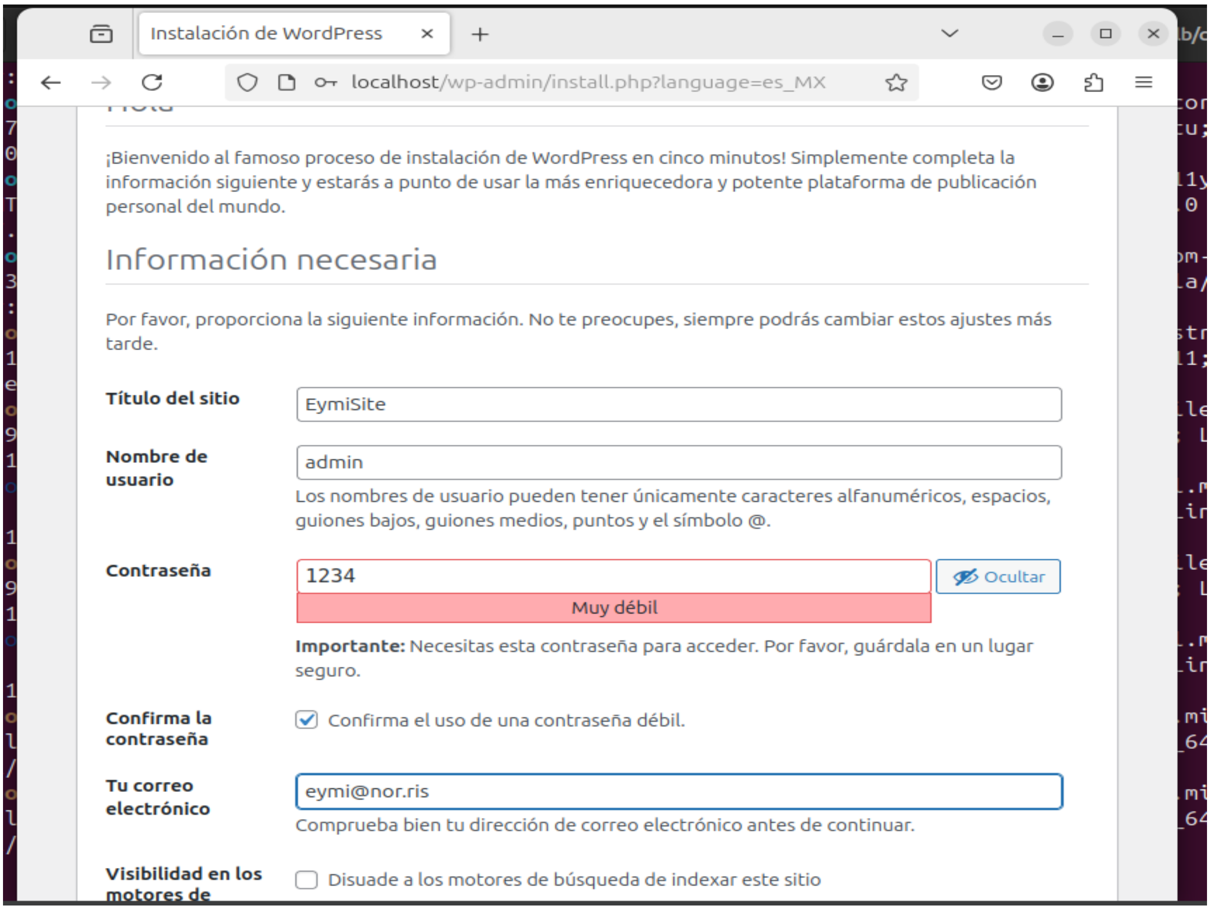
:

`

13. Configuramos en el wordpress usuario, contraseña y la db



14. Configuramos la instalación



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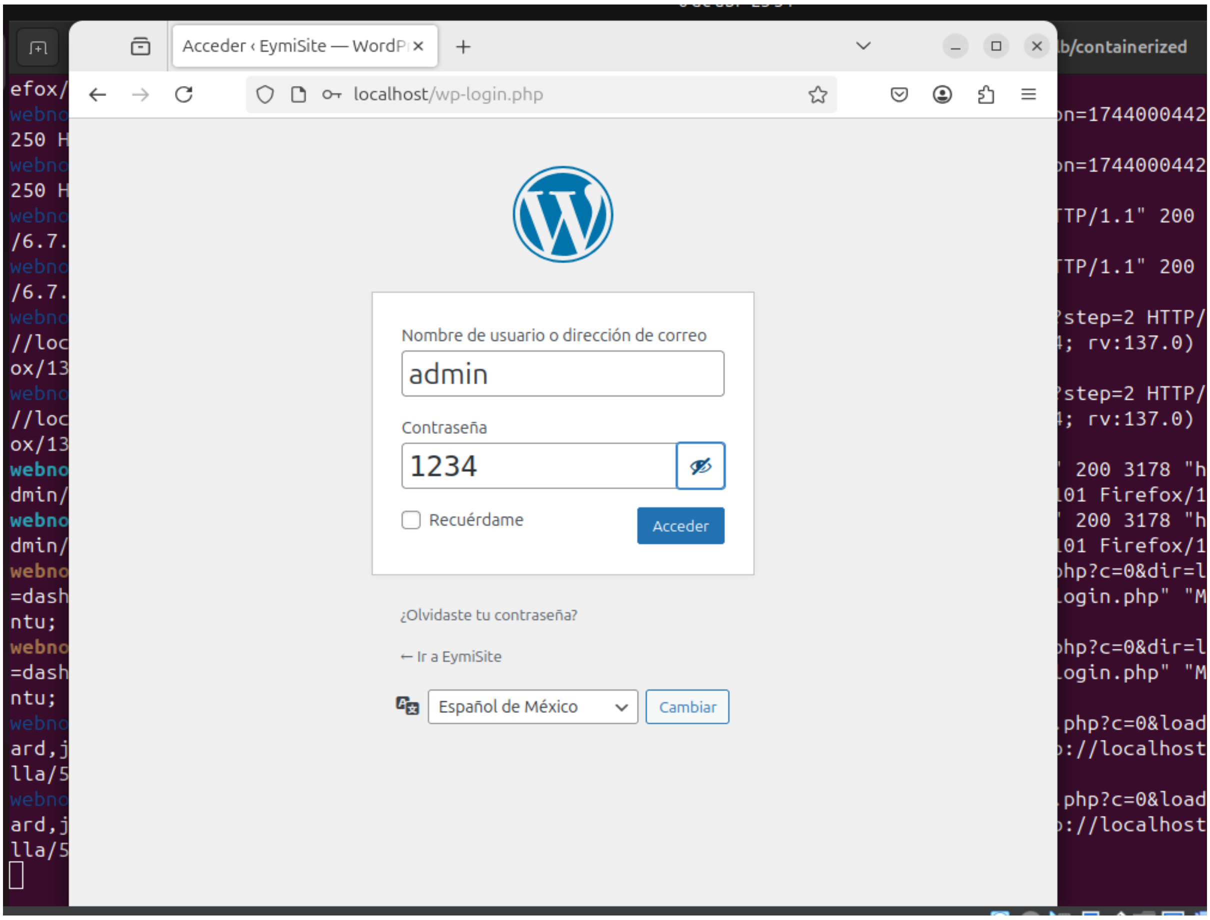
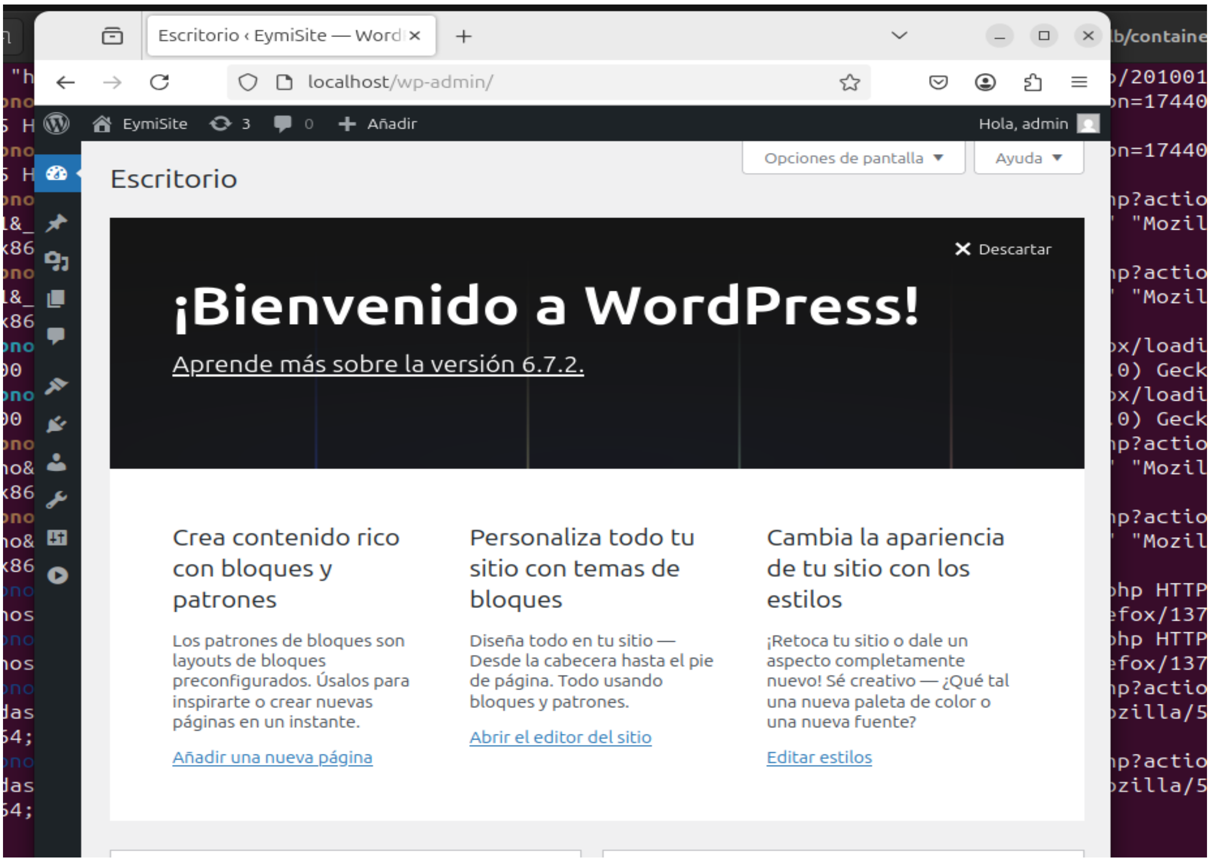
`

15. Nos logeamos





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PRUEBAS CON SIEGE

(Con base a usuarios)

1.

```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized# siege -c 15 -t
1m -i -b http://localhost
```

```
webnode1      | 172.18.0.110 - - [07/Apr/2025:04:53:16 +0000] "GET / HTTP/1.1" 2
-gnu) Siege/4.0.7"

{       "transactions":                    5185,
        "availability":                  100.00,
        "elapsed_time":                   59.86,
        "data_transferred":               18.44,
        "response_time":                   0.17,
        "transaction_rate":               86.62,
        "throughput":                      0.31,
        "concurrency":                    14.86,
        "successful_transactions":         5185,
        "failed_transactions":                0,
        "longest_transaction":             1.30,
        "shortest_transaction":            0.00
}
```

2.

```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized# siege -c 50 -t
1m -i -b http://localhost
```

```
-gnu) Siege/4.0.7"

{       "transactions":                    4988,
        "availability":                  100.00,
        "elapsed_time":                   59.54,
        "data_transferred":               17.73,
        "response_time":                   0.58,
        "transaction_rate":               83.78,
        "throughput":                      0.30,
        "concurrency":                    49.00,
        "successful_transactions":         4988,
        "failed_transactions":                0,
        "longest_transaction":             5.95,
        "shortest_transaction":            0.00
}
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized# webnode2    | 1
```

3.

```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized# siege -c 200 -t
 1m -i -b http://localhost
```

`


```
{       "transactions":                    2952,
        "availability":                   73.87,
        "elapsed_time":                   39.10,
        "data_transferred":               12.91,
        "response_time":                   2.48,
        "transaction_rate":               75.50,
        "throughput":                      0.33,
        "concurrency":                   187.35,
        "successful_transactions":         2952,
        "failed_transactions":             1044,
        "longest_transaction":            18.30,
        "shortest_transaction":            0.00
}
```


4.

```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized# siege -c 255 -t
 1m -i -b http://localhost
```


```
{       "transactions":                    3352,
        "availability":                   75.61,
        "elapsed_time":                   42.61,
        "data_transferred":               14.42,
        "response_time":                   3.02,
        "transaction_rate":               78.67,
        "throughput":                      0.34,
        "concurrency":                   237.84,
        "successful_transactions":         3352,
        "failed_transactions":             1081,
        "longest_transaction":            19.97,
        "shortest_transaction":            0.00
}
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-w
```


De acuerdo con la capacidad que tengo en el servidor, la cantidad de nodos web y nodos de bd podemos determinar la capacidad que puede soportar de acuerdo con las pruebas realizadas con el comando siege. Se realizó pruebas con diferentes cantidades de usuarios y a mayor cantidad de usuarios el portal web empieza a fallar con las transacciones realizadas. Con estos resultados podemos determinar cuál es el límite de nuestro sitio web.

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PRUEBAS CON AB

(simulando la cantidad de request)

1.

```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized# ab -n 1000 -c 2
00  http://localhost/
```

```
webnode3    | 172.18.0.110 - - [07/Apr/2025:05:24:14 +0000] "GET / HTTP/1.0" 200 49926 "-" "ApacheBench/2.3"
Concurrency Level:      200
Time taken for tests:   24.846 seconds
Complete requests:      1000
Failed requests:        415
   (Connect: 0, Receive: 0, Length: 415, Exceptions: 0)
Non-2xx responses:      415
Total transferred:      30323980 bytes
HTML transferred:       30065230 bytes
Requests per second:    40.25 [#/sec] (mean)
Time per request:       4969.173 [ms] (mean)
Time per request:       24.846 [ms] (mean, across all concurrent requests)
Transfer rate:          1191.88 [Kbytes/sec] received
webnode3    | 172.18.0.110 - - [07/Apr/2025:05:24:14 +0000] "GET / HTTP/1.0" 200 49926 "-" "ApacheBench/2.3"

Connection Times (ms)
              min  mean[+/-sd] median   max
Connect:        0    2   4.9      0      21
Processing:    23 4584 4170.7   3605   15404
Waiting:       23 4353 3940.2   3338   15223
Total:         23 4586 4170.7   3614   15404

Percentage of the requests served within a certain time (ms)
  50%   3614
  66%   6742
  75%   8341
  80%   9063
  90%  10748
  95%  11668
  98%  13068
  99%  13771
 100%  15404 (longest request)
```

2.

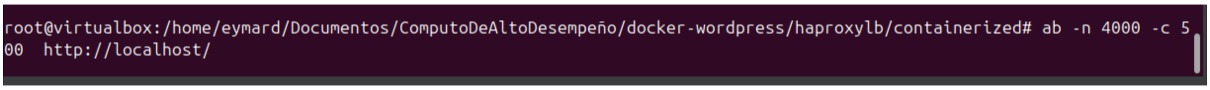
```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized# ab -n 4000 -c 1
000  http://localhost/
```

```
Concurrency Level:      1000
Time taken for tests:   40.487 seconds
Complete requests:      4000
Failed requests:        3188
   (Connect: 0, Receive: 0, Length: 3188, Exceptions: 0)
Non-2xx responses:      3188
Total transferred:      49192656 bytes
HTML transferred:       48081256 bytes
Requests per second:    98.80 [#/sec] (mean)
Time per request:       10121.848 [ms] (mean)
Time per request:       10.122 [ms] (mean, across all concurrent requests)
Transfer rate:          1186.53 [Kbytes/sec] received
webnode2    | 172.18.0.110 - - [07/Apr/2025:05:30:19 +0000] "GET / HTTP/1.0" 200 49926 "-" "ApacheBench/2.3"
webnode2    | 172.18.0.110 - - [07/Apr/2025:05:30:19 +0000] "GET / HTTP/1.0" 200 49926 "-" "ApacheBench/2.3"

Connection Times (ms)
              min  mean[+/-sd] median   max
Connect:        0   16  27.5      0      90
Processing:   223 8908 6577.1   5492   36888
Waiting:       97 8808 6432.3   5486   36343
Total:        223 8924 6591.3   5494   36936

Percentage of the requests served within a certain time (ms)
  50%   5494
  66%   7924
  75%  13285
  80%  15756
  90%  19177
  95%  20987
  98%  27191
  99%  30113
 100%  36936 (longest request)
```

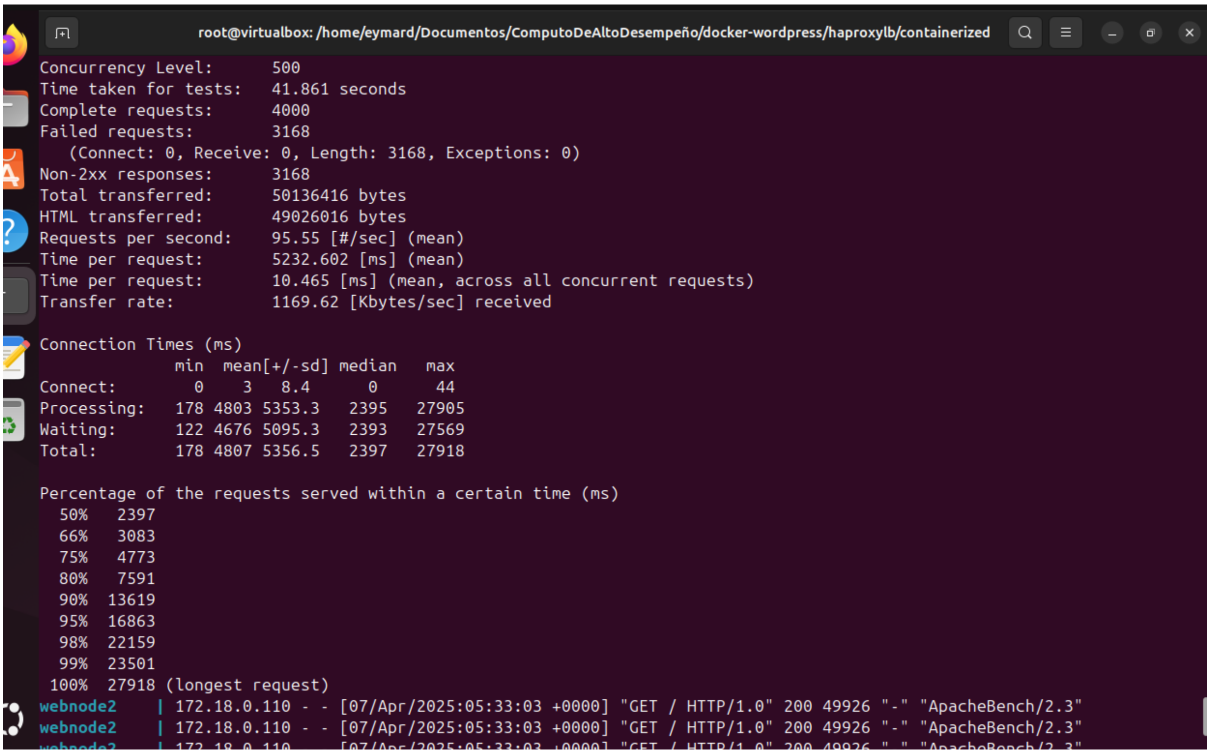
:

`

3.



```
root@virtualbox:/home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized# ab -n 4000 -c 5
00  http://localhost/
```

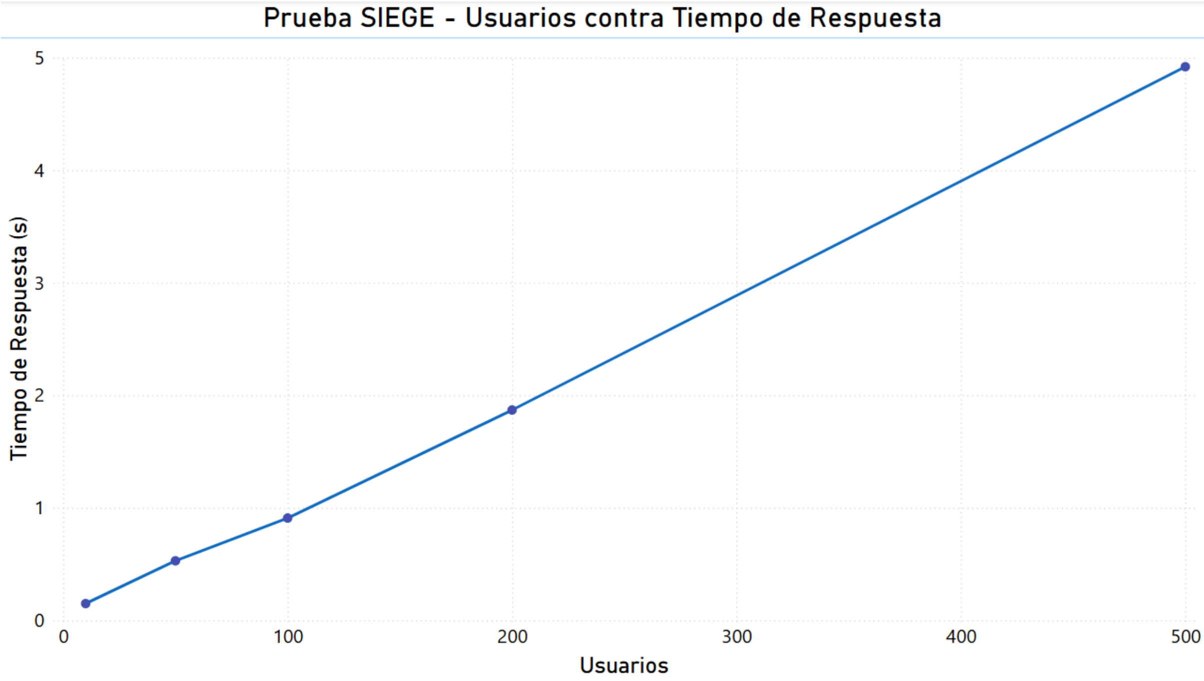


```
root@virtualbox: /home/eymard/Documentos/ComputoDeAltoDesempeño/docker-wordpress/haproxylb/containerized
Concurrency Level:      500
Time taken for tests:   41.861 seconds
Complete requests:      4000
Failed requests:        3168
   (Connect: 0, Receive: 0, Length: 3168, Exceptions: 0)
Non-2xx responses:      3168
Total transferred:      50136416 bytes
HTML transferred:       49026016 bytes
Requests per second:    95.55 [#/sec] (mean)
Time per request:       5232.602 [ms] (mean)
Time per request:       10.465 [ms] (mean, across all concurrent requests)
Transfer rate:          1169.62 [Kbytes/sec] received

Connection Times (ms)
              min  mean[+/-sd] median   max
Connect:        0    3   8.4      0      44
Processing:   178 4803 5353.3   2395   27905
Waiting:      122 4676 5095.3   2393   27569
Total:        178 4807 5356.5   2397   27918

Percentage of the requests served within a certain time (ms)
  50%   2397
  66%   3083
  75%   4773
  80%   7591
  90%  13619
  95%  16863
  98%  22159
  99%  23501
 100%  27918 (longest request)
webnode2     | 172.18.0.110 - - [07/Apr/2025:05:33:03 +0000] "GET / HTTP/1.0" 200 49926 "-" "ApacheBench/2.3"
webnode2     | 172.18.0.110 - - [07/Apr/2025:05:33:03 +0000] "GET / HTTP/1.0" 200 49926 "-" "ApacheBench/2.3"
```

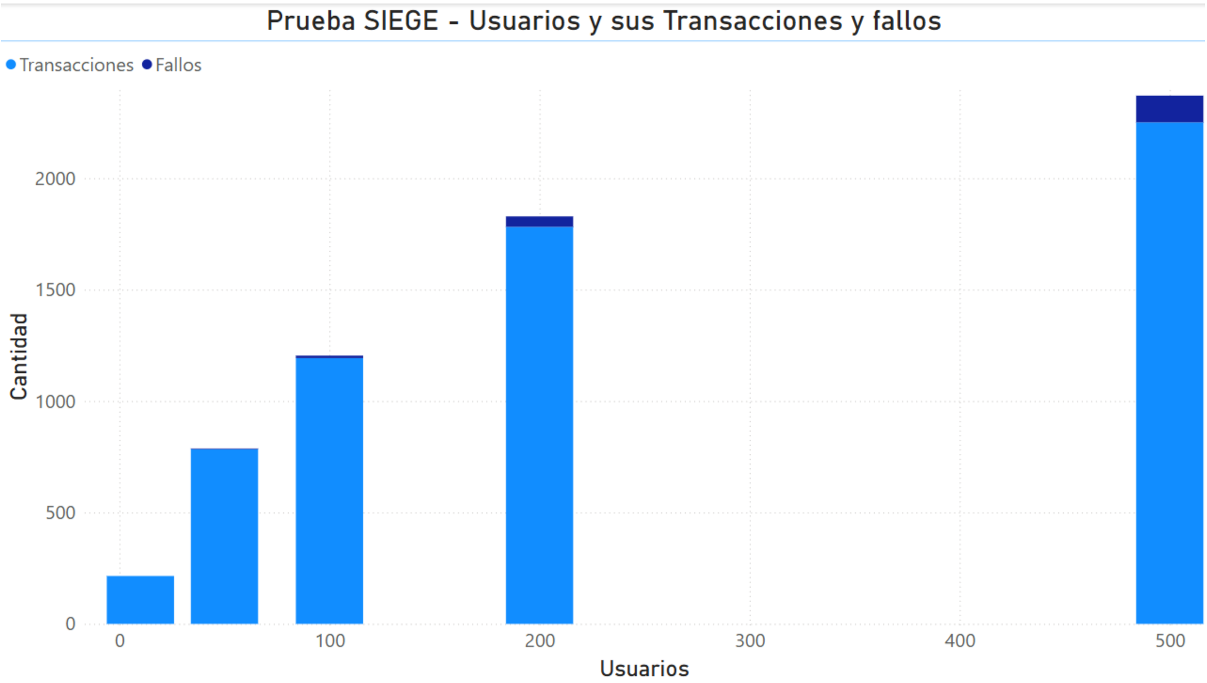
:

`

DASHBOARDS



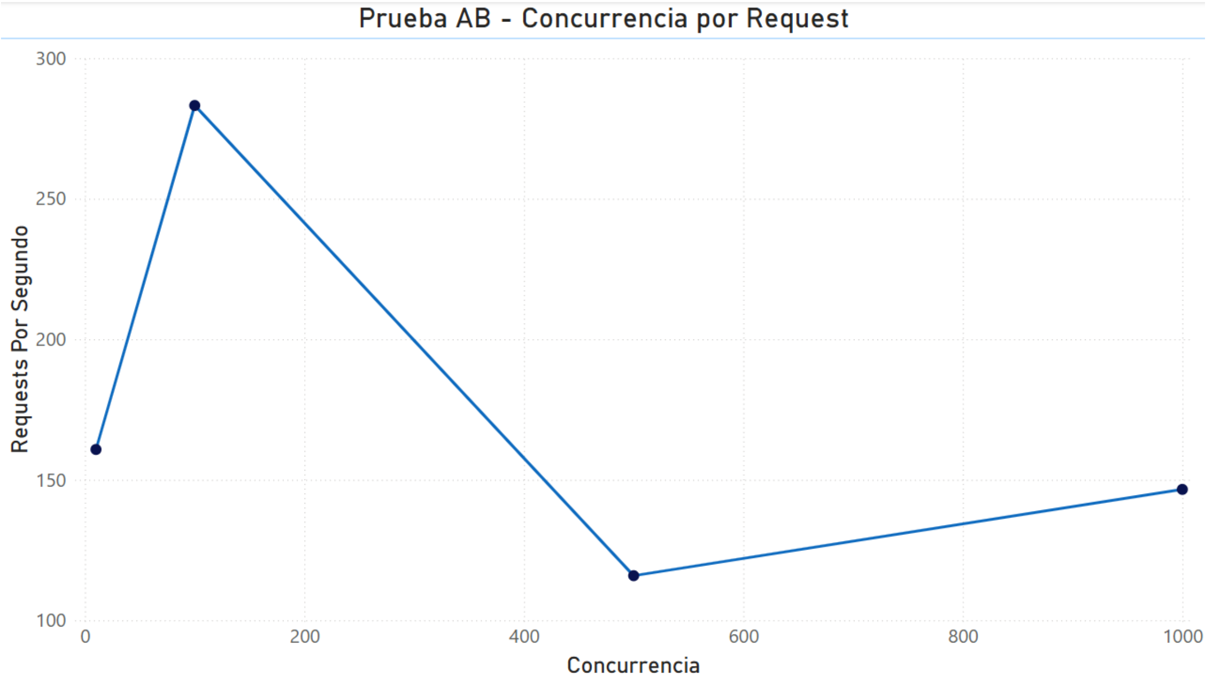
Prueba SIEGE - Usuarios contra Tiempo de Respuesta
Tiempo de Respuesta (s)
Usuarios
0
1
2
3
4
5
0
100
200
300
400
500

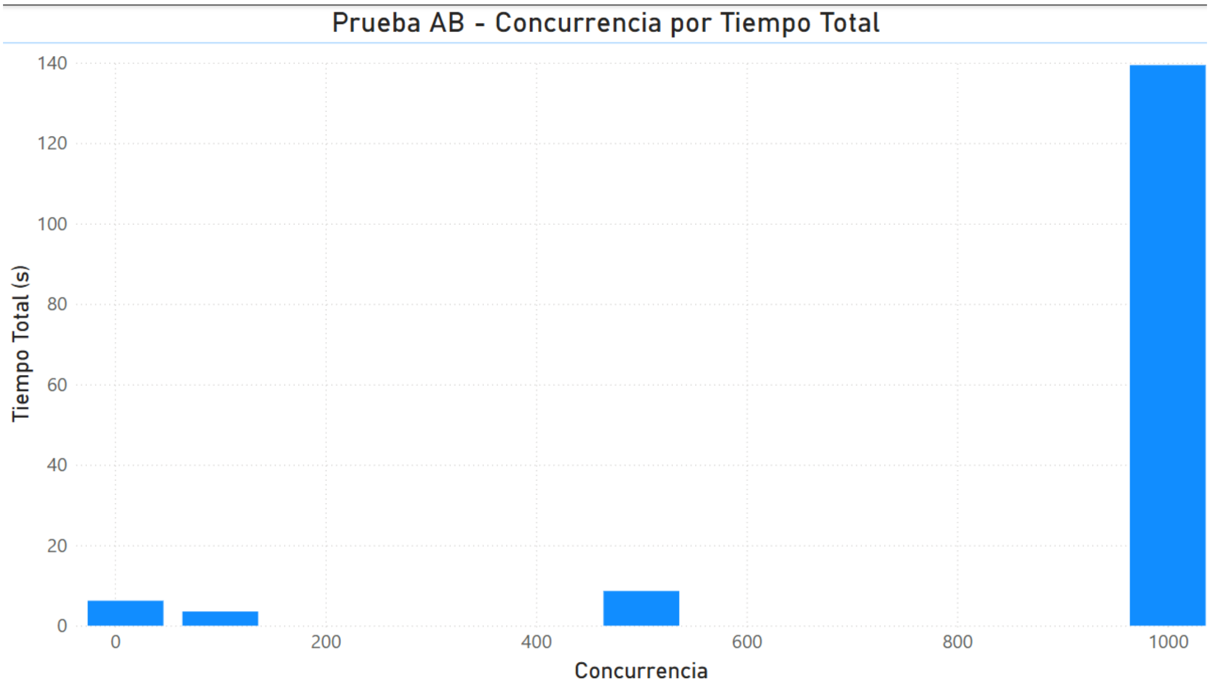


Prueba SIEGE - Usuarios y sus Transacciones y fallos
Transacciones
Fallos
Cantidad
Usuarios
0
500
1000
1500
2000
0
100
200
300
400
500

:

`


Prueba AB - Concurrencia por Request
Requests Por Segundo
300
250
200
150
100
0
200
400
600
800
1000
Concurrencia


Prueba AB - Concurrencia por Tiempo Total
Tiempo Total (s)
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
Concurrencia

:

`

CONCLUSIONES

El clúster HA tanto para Wordpress como MariaDB se completó, ya que se logró levantar los servicios correctamente, poder replicar las bd de los nodos y distribuir las cargas con un balanceador.

Mis pruebas de rendimiento con siege y ab me permitieron visualizar que tanto responde mi servidor a diferentes pruebas de niveles de carga, mientras que por un lado responde bien a cargas moderadas, por otro lado cuando es mayor, empieza a bajar su rendimiento.

Con esto pude saber igual cuál es el límite de quiebre y aprender en mejorar tal vez la infraestructura.

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